Christian Garry

Machine Learning Researcher | Stochastic Control · Deep Learning · Multi-Agent RL
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Research

Deep BSDEs for Mean-Field Market Making — Durham, United Kingdom
Durham University — 2026 – MSc Dissertation

- Built deep backward-SDE-with-jumps solvers (PyTorch) for a competitive mean-field market-making game (Cont–Xiong); matched the exact solution to 0.135% spread error and scaled to ~5,000 agents.
- **Tinkered architectures into working models:** diagnosed and fixed three failure modes that silently kill the mean-field signal (generator bypass, BatchNorm erasure, DeepSets collapse); validated over a 26-experiment ablation battery, with the error theory machine-checked in Lean 4 / Mathlib (~36k LOC).
- 20-seed multi-agent-RL (MADDPG) study: learned spreads reach **supra-cartel** (t=3.22, p=0.0022) — collusion is a property of the learning dynamics, not the market equilibrium.

Silicon Carbide JFET CPU — Durham, United Kingdom
MEng Dissertation (84%) — Oct 2023 – Apr 2024

- Designed and simulated a 4-bit CPU in SiC JFET logic (LTspice) for extreme-temperature/radiation computing.
- Built the full verification toolchain — C++ compiler, assembler, Python automation, emulator-parity + waveform-to-state checks.

Education

MSc Scientific Computing & Data Analysis (AI for Engineering) — Durham, United Kingdom
Durham University — Sep 2025 – Exp. Sep 2026

- Tracking Distinction — average ~86% (95% performance/GPU, 94% machine learning & statistics, ~89% advanced Bayesian ML).
- **Focus:** Bayesian inference (MCMC, Gaussian processes), deep learning (CNNs, Transformers, diffusion), convex optimisation (duality, KKT), GPU/HPC. Python, C, R.

University of Illinois Urbana-Champaign — Mathematics Modules (NetMath, online) — Apr 2026 – Dec 2026

- **MATH 314** Higher Mathematics *(in progress, 100%)* & **MATH 447** Real Analysis *(upcoming)* — proof techniques, metric spaces, compactness, Riemann integration.

MEng Electronic Engineering — Durham, United Kingdom
Durham University — Sep 2020 – Jun 2024

- Upper Second-Class Honours (2:1). Dissertation: *Silicon Carbide JFET CPU* (see Research).

Experience

Graduate Communications Engineer — Hebburn, United Kingdom
Siemens PLC — Sep 2024 – Present

- Built a Python RAG platform from scratch (local LLMs, vector search) over large technical corpora — cut engineer time-to-answer by >90%.
- Implemented C/C++ communication components (TCP/IP, serial) for digital-twin simulation.

Technical Skills

Deep learning (PyTorch, deep-BSDE/FBSNN, CNNs/Transformers) • architecture & feature debugging • multi-agent RL (MADDPG) • Bayesian inference (MCMC, Gaussian processes) • stochastic control (HJB↔BSDE, mean-field games) • hypothesis testing with multiple-comparison correction • GPU/CUDA & HPC • Python, C++, C, R • Lean 4 / Mathlib